



The Hydrosphere — FULL MIND MAP

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Chapter 20_The_Hydrosphere

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Ocean Mean Temperature (OMT)

- 26°C isotherm generally 50-100m depth pe mili hai — Jan-March SW Indian Ocean mein average depth ~59m hai (129m NAHI, Statement 1 FALSE). **OMT SST se better predictor** hai India ke monsoon rainfall (long-term mean 887.5mm se kam/zyada) predict karne ke liye — Statement 2 **TRUE**. [Q1]

Earth's Surface — Water Coverage

- Earth ki surface ka ~70.9% (two-thirds se zyada) water se covered hai (~361.9 million sq km) — sirf ~29.1% land hai (~148.17 million sq km). [Q2]

Earth's Centre — Nearest Point at Sea Level

- Earth **oblate spheroid** hai (polar radius equatorial radius se kam) — sea level pe **North Pole** Earth ke centre ke sabse close hai (Equator se zyada close). [Q3]

Datum Line

- **Datum Line** = sea level ke reference mein **horizontal line** hai, jisse heights aur depths measure hoti hain. [Q4]

Pacific Ocean — Largest

- **Pacific Ocean** = hydrosphere ka largest part (~155.6 million sq km) — Atlantic (~76.8M), Indian (~68.6M), Southern (~20.3M) se bada. [Q5]

Mid-Oceanic Ridge

- **Mid-Oceanic Ridge** (~64,000-65,000km lambi, 2,000-2,400km chaudi) — Arctic, Atlantic, Indian, Pacific oceans se guzarti hai, Earth ki sabse badi continuous geological features mein se ek hai — North/South Atlantic se hoke Indian Ocean se South Pacific mein enter karti hai, Australia-Antarctica ke beech tak extend hoti hai. [Q6]

Ninety East Ridge

- **Ninety East Ridge (90°E Ridge)** = **Indian Ocean** mein hai — ~5,000km lambi, ~34°S se 10°N tak, average width ~200km, 90°E longitude ke roughly parallel. [Q7]

Gravitational Water — Tension

- **Gravitational Water** (soil macropores mein) ka tension **1/3 atmosphere se kam** hota hai (ya zero) — gravity ki wajah se easily downward move karta hai. [Q8]

Freshwater Distribution

- Earth ka total water: ~96.5% saline ocean water, ~2.5% freshwater. Human-use ke liye directly available water Earth ke total water ka **1% se kam** hai (Statement 1 TRUE); total freshwater ka **~68.7% glaciers/polar ice** mein hai, 95% NAHI (Statement 2 galat number, sahi ~68.7% hai) — is question mein sirf Statement 1 correct answer hai. [Q9]
- Rivers/lakes ka water groundwater se **KAM** hai (Statement 1 FALSE); polar ice/glaciers ka water groundwater se **ZYADA** hai (Statement 2 TRUE). [Q10]
- Planet pe sabse zyada freshwater **Glaciers aur mountains (permanent ice/snow, ~68.7%)** mein hai — Groundwater ~30.1%, lakes/rivers bahut kam. [Q11]
- Glaciers/ice ke baad, remaining freshwater ka largest proportion **Groundwater** (~30.1%) mein hai — freshwater lakes ~0.26%, atmosphere ~0.04%, rivers ~0.006%. [Q12]
- World water sources descending order: **Oceans (96.5%) > Ice/Glaciers (1.74%) > Groundwater (1.7%) > Rivers (0.0002%)**. [Q13]

⚠ EXAM TRAP

- Total freshwater ka glaciers/polar-ice share **~68.7%** hai (95% NAHI) — exact percentage yaad rakho, alag questions mein alag context mein use hota hai. [Q9]

Oceanic Floor — Most Extensive Part

- **Deep Sea (Abyssal) Plains** = oceanic floor ka sabse extensive/broad part hain (Ocean Deepes, Oceanic Slope, Continental Shelves se zyada area cover karte hain) — several thousand metres depth, fine sediments se covered. [Q14]

Red Sea

- **Red Sea = Axial Trough** ka example hai (Africa-Arabia ke beech, magnetic anomaly bhi identified) — Suez Canal se commercial importance badhi (North Africa/Europe ko Asia se connect karta hai). [Q15]

Ocean Ridges/Plateaus/Basins — Location Mismatch

- Correct pairs: Cocoa Ridge-Pacific, Walvis Ridge-Atlantic, Agulhas Basin-Indian Ocean — **Telegraph Plateau Atlantic Ocean mein hai, Indian Ocean NAHI** (yeh GALAT combination hai). [Q16]

Indian Ocean — Ancient Name aur Equator

- Indian Ocean ko ancient Greeks '**Erythraean Sea**' kehte the (Red Sea, Gulf of Aden, Arabian Sea, northern Indian Ocean ke liye collectively) — Statement 1 TRUE; Indian Ocean **Equator ke dono taraf phaila hai** (~30°N se 60°S) — Statement 2 bhi TRUE. [Q17]